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# New paleomagnetic and geochronologic results from Ethiopian Afar: Block rotations linked to rift overlap and propagation and determination of a ~2 Ma reference pole for stable Africa

**Tesfaye Kidane**
Laboratoires de Paléomagnétisme et de Tectonique, Institut de Physique du Globe de Paris,
Paris, France

Department of Geology and Geophysics, University of Addis-Ababa,
Addis-Ababa, Ethiopia

**Vincent Courtillot and Isabelle Manighetti**
Laboratoires de Paléomagnétisme et de Tectonique, Institut de Physique du Globe de Paris,
Paris, France

**Laurence Audin**
Laboratoires de Paléomagnétisme et de Tectonique, Institut de Physique du Globe de Paris,
Paris, France

Institut de Recherche pour le Devoloppement, Université Paul Sabatier, Géologie,
Toulouse, France

**Pierre Lahitte**
Laboratoire de Géochronologie Multi-Techniques, Université Paris Sud–Institut de Physique du Globe de Paris, Sciences de la Terre, Université Paris XI-Paris Sud,
Orsay, France

**Xavier Quidelleur**
Laboratoires de Paléomagnétisme et de Tectonique, Institut de Physique du Globe de Paris,
Paris, France

Laboratoire de Géochronologie Multi-Techniques, Université Paris Sud–Institut de Physique du Globe de Paris, Sciences de la Terre, Université Paris XI-Paris Sud, Orsay, France

**Pierre-Yves Gillot**
Laboratoire de Géochronologie Multi-Techniques, Université Paris Sud–Institut de Physique du Globe de Paris, Sciences de la Terre, Université Paris XI-Paris Sud, Orsay, France

**Yves Gallet**
Laboratoires de Paléomagnétisme et de Tectonique, Institut de Physique du Globe de Paris,
Paris, France

**Julie Carlut**
Laboratoire de Géologie, Ecole Normale Supérieure,
Paris, France

**Tigistu Haile**
Department of Geology and Geophysics, University of Addis-Ababa,
Addis-Ababa, Ethiopia

## Abstract

[1] Joint French–Ethiopian field trips in 1995–1996 yield new geochronologic and paleomagnetic data, which significantly expand our knowledge of the recent magmatic and tectonic history of the Afar depression. Twenty-four new K-Ar ages range from 0.6 to 3.3 Ma. There is quite good agreement between magnetic polarities and Geomagnetic Polarity Timescale (GPTS). Eight age determinations with uncertainty less than 50 kyr can be used in future reassessments of the GPTS (upper and lower Olduvai/Matuyama reversals and Reunion and Mammoth subchrons). Paleomagnetic analysis of 865 cores from 133 sites confirms that low-Ti magnetites are the main carrier of the Characteristic Remanent Magnetization (ChRM). A positive tilt test (based on two subgroups with 63 and 23 sites, respectively) confirms that this ChRM is likely the primary magnetization. The main paleomagnetic results can be summarized as follows. A ~2 Ma reference pole for stable Africa is determined based on 26 sites located on either side of the northern termination of the East African rift. It is located at $\lambda = 87.2°$N, $\Phi = 217.1°$E ($A_{95} = 4°$). A $4.6 \pm 1.8°$ ($2\sigma$) inclination shallowing is identified within a population of 231 stratoid lava flows, consistent with a global axial quadrupole of $6 \pm 2\%$ of the axial dipole. Combined with earlier data of *Acton et al.* [2000], our new data allow mean paleomagnetic field directions to be determined for five individual, fault-bounded blocks previously identified by tectonic analysis within central Afar. These all have suffered negligible rotations about vertical axes since emplacement of the lava. This contrasts with the

significant rotations previously uncovered to the east in Djiboutian Afar for three major individual blocks. Taken altogether, the declination differences with respect to reference directions are 2 ± 4° for central Afar and 13 ± 4° for eastern Afar, consistent with the model of *Manighetti et al.* [2001a]. It appears that in the last ~3 Ma the Afar depression was extensively floored by trap-like basalts, which were deformed by a single but complex physical (tectonic) process, combining diffuse extension, rift localization, propagation, jumps and overlap, and bookshelf faulting.

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**Keywords:** Afar tectonics, block rotation, rift propagation and overlap, reference pole, polarity timescale, inclination anomaly.

**Index Terms:** 1525 Geomagnetism and Paleomagnetism: Paleomagnetism applied to tectonics (regional, global); 1035 Geochemistry: Geochronology; 8109 Tectonophysics: Continental tectonics—extensional (0905); 3040 Marine Geology and Geophysics: Plate tectonics (8150, 8155, 8157, 8158); 9604 Information Related to Geologic Time: Cenozoic.

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